

Stuff Inserts into a Carton SOP

One episode stuffs, closes, and seals one carton. The carton is already standing open and square on the stuffing spot with all four top flaps up. It takes one flyer laid flat and one sample stood upright, is held still while both inserts are confirmed, has its four top flaps folded in, is sealed with one tape strip along the top seam, and slides left onto the finished spot.

The order never changes: flyer, sample, confirm, close, seal, slide out. No sample goes in before the flyer is flat on the carton floor, no flap folds before both inserts are confirmed, and no strip comes out of the tape dispenser before all four flaps are in.

The right gripper takes the flyer, the sample, and the presented tape strip, folds three of the four top flaps, makes the seal press, and supplies the push on the final slide. The left gripper folds the left flap, steadies the carton for every insert, every fold, and the press, and guides the slide. Nothing is turned in the air, and a gripper opens only over the spot the thing in it is going to.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the carton standing on the stuffing spot at the center, the finished spot on its left, and the flyer spot, sample spot, tape dispenser, and tape scrap rest on its right.
3. The stuffing spot is visible from above, so the flyer face, the standing sample, every flap fold, and the strip landing on the seam can all be seen.
4. Both arms are at home with grippers open.
5. The tabletop is clear of anything but the zones listed below.
6. Both arms reach the stuffing spot and the finished spot without crossing or reaching a joint limit. The right arm also reaches the flyer spot, sample spot, tape dispenser, and tape scrap rest. Both grippers can travel beside the carton through the slide without touching it anywhere but its walls.
7. The straight path from the stuffing spot to the finished spot is bare.

Materials checklist

1. One small **carton** stands square on the **stuffing spot** with its **near wall** toward the front edge. Its bottom is already taped, it is empty, and all four **top flaps** stand up and clear of the opening.

2. One **flyer**, a flat printed sheet, lies on the **flyer spot** at the front right, printed face up. It fits flat on the carton floor with room to spare.
3. One **sample**, a capped unit that stands on its own, sits upright on the **sample spot** at the right edge, cap up, with clear space around it so a gripper can take it from above.
4. One tabletop **tape dispenser** stands to the right of the stuffing spot. It contains enough tape for the episode and any retries and automatically presents one strip at a time cut to length, adhesive face down, with a stiff **grab tab** accessible to the right gripper. Its stock is not counted during the episode.
5. One bare **tape scrap rest** sits beside the dispenser for a strip that must be scrapped.
6. The stuffing spot, the finished spot, and the path between them are bare, clean, and dry.

Workspace layout

- **Stuffing spot:** the center of the table, where the carton stands and every insert, fold, and seal happens. It holds the carton and nothing else.
- **Finished spot:** the clear unmarked area immediately left of the stuffing spot. The sealed carton stands here at the end of the episode.
- **Tape dispenser:** right of the stuffing spot, presenting one cut strip at a time by its grab tab.
- **Tape scrap rest:** beside the dispenser, where a damaged or badly laid strip is placed.
- **Flyer spot:** front right, one flyer lying flat and printed face up.
- **Sample spot:** right edge, one sample standing upright with its cap up.

Everything the right gripper fetches is staged on the right. The stuffing spot and the finished spot are shared, and both grippers reach the carton in either position.

Vocabulary

- **Arm assignment:** the right gripper fetches the flyer, the sample, and the presented tape strip, folds the near, far, and right top flaps, presses the seal, and supplies the push on the slide. The left gripper folds the left top flap, holds the carton for every insert and fold, and guides the slide. Neither gripper takes over the other's work.
- **Carton:** the small open box standing on its taped bottom. Its **near wall** faces the front edge and its **far wall** faces away.
- **Top flaps:** the four flaps standing up around the carton opening, named for the wall they hinge from: **near flap**, **far flap**, **left flap**, **right flap**.
- **Top seam:** the line down the middle of the closed top where the left flap and the right flap meet. It runs near to far and it is the only place a strip goes.
- **Left slide: Unvalidated at the station.** Both grippers closed before contact. The right gripper presses the middle of the carton's right wall toward the finished spot while the left gripper travels against the left wall as a guide. The carton keeps its facing and its base stays flat on the table.

- **Flyer:** the flat printed sheet. It lies on the carton floor **printed face up** with no corner folded under and no corner riding up a wall.
- **Sample:** the capped unit that stands on its own. It goes in **upright, cap up**, standing on the carton floor beside the flyer and never on top of it.
- **Tape dispenser: Unvalidated at the station.** The tabletop machine automatically advances and cuts one strip ready to use after the presented strip is removed. The grippers do not operate its cutter or reach inside it.
- **Strip:** one cut length of tape presented by the dispenser. Its **grab tab** is the stiff end carrying no adhesive, and it is the only part of a strip a gripper touches. A strip touched anywhere else is scrapped.
- **Hangs straight:** the strip dangling free and flat from the lifted grab tab, not curled, twisted, or stuck to itself.
- **Laid along the seam:** the far end of the hanging strip set down on the far end of the top seam, then drawn toward the front edge along the seam so the strip settles onto it behind the gripper, with the tab released past the near end.
- **Flat hold:** the left gripper closed and resting on the carton's near wall to keep it from shifting while the right gripper works. It goes on before the right gripper moves and comes off after the right gripper is clear.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Lower in:** carrying level over the opening, lowering straight down until the thing is resting on the carton floor, and releasing once it is resting. Nothing is dropped in from above the rim.
- **Settled:** the thing stays put for 2 seconds after the gripper lifts clear, with nothing sliding, toppling, springing open, or hanging over an edge.
- **Release point:** inside the carton, on the tape scrap rest, or on the applied strip after it is resting on the seam. A gripper opens nowhere else. The carton is moved with closed grippers and is never held or released.

Steps

Steps 1 to 5 finish the carton, in that order. Step 6 ends the episode once the sealed carton stands on the finished spot.

Step 1: Load the flyer and the sample

Goal: the flyer lying flat on the carton floor printed face up, and the sample standing upright beside it.

The **left gripper** takes a **flat hold** on the carton's near wall before the first insert and keeps it through both inserts.

1.1 The flyer

- The **right gripper** closes on the near edge of the **flyer** on the flyer spot, lifts it straight up, carries it level over the carton opening, lowers it in until it is resting on the carton floor, and releases.
- The flyer goes in printed face up, the way it lay on its spot. It is not turned, folded, or curled on the way in.

1.2 The sample

- The **right gripper** closes on the **sample** on the sample spot, lifts it straight up, carries it level and upright over the carton opening, lowers it in until it is standing on the carton floor beside the flyer, and releases once it is standing.
- The sample stays upright the whole way and never rides on top of the flyer.

The **left gripper** releases the flat hold once the right gripper is clear of the opening.

Take one thing per trip. The flyer is fully in and released before the sample is picked.

Check: the flyer lies flat on the carton floor printed face up with no corner folded under or riding a wall, and the sample stands upright beside it, cap up, on the carton floor. If the flyer landed folded or the sample landed leaning, lift it clear and put it in again.

Step 2: Confirm the orientation of both inserts

Goal: both inserts seen and held still, so the flyer face and the standing sample can be read from above.

- Lift both grippers clear of the carton opening.
- Leave the open carton untouched for 2 seconds with the stuffing spot in frame. Nothing is held, nudged, or moved during the hold.
- If the flyer is face down, folded, or riding a wall, or the sample is leaning, lying down, or standing on the flyer, correct it now by lifting it clear and putting it in again, then hold for 2 seconds again.

Nothing is folded, sealed, or fetched until this hold is done.

Check: during the hold the flyer is flat and printed face up on the carton floor and the sample stands upright with its cap up, both visible from above, and both grippers are clear of the opening.

Step 3: Close the four top flaps

Goal: all four top flaps folded in flat, with the top seam running near to far down the middle.

Fold in this order and no other. Every fold is a fold down onto the flaps already in, never a press on the carton contents.

3.1 Near flap, then far flap

- The **left gripper** takes a **flat hold** on the carton's near wall and keeps it for the whole step.
- The **right gripper** pinches the free edge of the **near flap**, folds it away from the front edge down over the opening, and releases.
- The **right gripper** pinches the free edge of the **far flap**, folds it toward the front edge down over the near flap, and releases.

Expected state: the opening covered near to far, the left and right flaps still standing up.

3.2 Left flap, then right flap

- The **left gripper** lifts off the near wall, pinches the free edge of the **left flap**, folds it right down over the flaps already in until it reaches the middle, releases, and goes back to its flat hold on the near wall.
- The **right gripper** pinches the free edge of the **right flap**, folds it left down over the flaps already in until its edge meets the left flap along the middle, and releases.

Check: all four flaps lie flat and down, no flap edge stands proud or is trapped under another flap, and the left and right flaps meet along the middle in one straight top seam running near to far. If a flap stands proud or the seam is off the middle, lift that flap and fold it again.

Step 4: Seal the top seam

Goal: one tape strip laid down the top seam and pressed flat, with the carton still square on the stuffing spot.

- The **left gripper** keeps its **flat hold** on the near wall for the whole seal.
- The **right gripper** pinches the **grab tab** of the strip presented by the **tape dispenser**, lifts it straight up until it **hangs straight**, and carries it level over the carton. The dispenser presents the next strip after this one clears it.
- Lay it **along the seam**: set the far end of the strip down on the far end of the top seam, draw the tab toward the front edge along the seam so the strip settles onto it behind the gripper, and release the tab once it lies past the near end.
- The **right gripper** presses along the strip from far to near in one stroke, without coming off the tape, until it lies flat with no lifted end and no bubble standing up.
- Release the right gripper, then the left.

A strip is touched only by its grab tab. If it folds onto itself before landing, place it on the tape scrap rest and take the next presented strip. If it lands off the seam or folds on the carton, peel it off, place it on the scrap rest, and take the next presented strip. A scrapped strip is never reused.

Check: one strip runs down the top seam from end to end, lies flat with no lifted end, and the flaps stay down when the carton moves. The dispenser has presented another strip, any

scrapped strip is on the tape scrap rest, and no strip is stuck to a gripper.

Step 5: Slide the sealed carton onto the finished spot

Goal: the sealed carton standing square on the finished spot, taped seam up.

- Close both grippers before either one touches the sealed carton.
- The **right gripper** presses against the middle of the carton's **right wall**. The **left gripper** rests against the middle of its **left wall** as a guide.
- Make one slow **left slide** until the carton stands centered on the **finished spot**. Keep its base flat on the table and its facing unchanged.
- Stop once it stands square, then lift both grippers straight clear.

Do not pinch, lift, tip, or turn the carton, and never move it by a flap.

Check: the sealed carton stands square on the finished spot with its taped seam up and its near wall still toward the front edge, and the stuffing spot is bare. If it stops short or starts to turn, slide it right back to the stuffing spot with the push on the left wall and the guide on the right wall, then repeat the left slide.

Step 6: Confirm the sealed carton and end the episode

- Confirm the carton stands on the finished spot, closed with all four flaps down and one strip along its top seam; and confirm the flyer spot and sample spot are empty, the stuffing spot is bare, and no strip is stuck to a gripper.
- Leave the scene untouched for 3 seconds with the finished spot, the stuffing spot, and the tape dispenser all in frame. Nothing is held or moved during the hold.
- Return both arms home, clear of the finished spot and the stuffing spot, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Peel the strip off the carton top, open all four flaps, and set the used strip aside for disposal.
2. Take the flyer and the sample out of the carton.
3. Check the carton for a crushed wall, a torn flap, or a bottom tape that has let go. Replace it if it will not stand square with its flaps standing up.
4. Stand the carton square on the stuffing spot, empty, bottom taped, near wall toward the front edge, all four top flaps standing up. Leave the path to the finished spot bare.
5. Lay the flyer flat on the flyer spot, printed face up. Replace it if it is creased or torn.
6. Stand the sample upright on the sample spot with its cap up. Replace it if it will not stand on its own or its cap is loose.
7. Remove all tape scraps from the scrap rest. Check that the tape dispenser contains enough tape for one episode and retries, cuts each strip to the full seam length, and presents the

next strip by a clean grab tab.

8. Wipe the stuffing spot, the finished spot, and the path between them, and confirm they are clean and dry. A dirty base or tabletop can make the carton catch or turn during the slide.
9. Vary which flyer face is up and which sample is used between episodes rather than running the same pairing every time. The count stays at one carton, one flyer, and one sample; tape comes from the dispenser on demand.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the sample goes in before the flyer, a flap folds before the confirm hold, a strip comes out of the dispenser before all four flaps are in, or the carton moves off the stuffing spot before the seam is sealed.
- **SOP rule broken:** Steps 1 to 5, the order never changes: flyer, sample, confirm, close, seal, slide out.
- **Coaching note:** flyer, sample, confirm, close, seal, slide out. Finish the step you are in before starting the next one.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes the flyer, the sample, or the strip, folds the near, far, or right flap, or makes the seal press; or the right gripper folds the left flap, or the push and guide roles are swapped during the slide.
- **SOP rule broken:** Steps 1 to 5, the right gripper fetches, performs the primary work, and supplies the push, while the left gripper holds, folds the left flap, and guides the slide.
- **Coaching note:** right fetches, works, and pushes; left holds, folds left, and guides.

Violation: Carton slid wrong

- **Visible cue:** the carton is pinched, lifted, tipped, turned, moved by a flap, or pushed without the other gripper guiding its side wall.
- **SOP rule broken:** Step 5, both grippers are closed before contact, the right gripper supplies the push, the left gripper guides, and the carton base stays flat on the table.
- **Coaching note:** let the table support the carton; right pushes and left guides.

Violation: More than one thing handled at once

- **Visible cue:** the flyer and the sample are carried in one trip, a second strip is taken while one is still held, or the carton is moved while a flyer, sample, or strip is still held.
- **SOP rule broken:** Steps 1, 4, and 5, take one thing per trip and set it down before picking the next.
- **Coaching note:** one flyer, one sample, one strip, one at a time.

Violation: Flyer put in wrong

- **Visible cue:** the flyer goes in printed face down, folded, curled, standing on edge, riding up a carton wall with a corner, or landing on top of the sample, and the flaps are folded over it anyway.
- **SOP rule broken:** Step 1.1, the flyer is lowered in printed face up and rests flat on the carton floor, and a bad landing is corrected by lifting it clear and putting it in again.
- **Coaching note:** face up, flat on the floor, first thing in. Everything after this is scored on it.

Violation: Sample put in wrong

- **Visible cue:** the sample goes in lying down, upside down, leaning against a wall, standing on the flyer, or it is turned or tipped between the lift and the set down.
- **SOP rule broken:** Step 1.2, the sample is carried level and upright and lowered in until it stands on the carton floor beside the flyer, cap up.
- **Coaching note:** upright the whole way, down beside the flyer, not on it.

Violation: Dropped in from above the rim

- **Visible cue:** a gripper opens above the carton opening and the flyer flutters in or the sample falls in, instead of the thing being lowered until it is resting.
- **SOP rule broken:** Step 1, lower the flyer and the sample straight down until each is resting on the carton floor, then release.
- **Coaching note:** take it all the way down, then let go. Nothing gets dropped in.

Violation: Carton not steadied

- **Visible cue:** the left gripper is off the near wall while the flyer or the sample goes in, while a flap is folded, or during the seal press, or the carton slides, rocks, or shifts under the right gripper.

- **SOP rule broken:** Steps 1, 3, and 4, the left gripper takes a flat hold on the near wall before the right gripper moves and keeps it until the right gripper is clear.
- **Coaching note:** the hold goes on first and comes off last. A loose carton ruins the seam.

Violation: Confirm hold skipped or worked through

- **Visible cue:** a flap is folded straight after the sample lands with no still scene between them, a gripper stays in or over the opening during the hold, or something is nudged, pushed down, or rearranged while the hold is meant to be running.
- **SOP rule broken:** Step 2, lift both grippers clear and leave the open carton untouched for 2 seconds with the stuffing spot in frame before anything is folded.
- **Coaching note:** hands off and count two. The reviewer has to see what went in.

Violation: Bad orientation left uncorrected

- **Visible cue:** the confirm hold runs with the flyer face down, folded, or riding a wall, or with the sample leaning, lying down, or on the flyer, and the flaps are folded anyway.
- **SOP rule broken:** Step 2, correct a wrong insert by lifting it clear and putting it in again, then hold for 2 seconds again.
- **Coaching note:** fix it and hold again. A confirm that confirms the wrong thing is worse than none.

Violation: Flaps folded in the wrong order

- **Visible cue:** a side flap goes in before the near and far flaps, the far flap goes before the near flap, a flap is trapped under one folded later, or a flap is left standing up when the strip comes out.
- **SOP rule broken:** Step 3, fold near, then far, then left, then right, each one down over the flaps already in.
- **Coaching note:** near, far, left, right. Every time, no exceptions.

Violation: Top seam off the middle or flaps proud

- **Visible cue:** the left and right flaps overlap heavily, leave a gap, or meet well off the middle, or a flap edge stands proud of the top, and the strip goes on anyway.
- **SOP rule broken:** Step 3.2, the left and right flaps fold in until they meet along the middle in one straight top seam running near to far, and a proud or crooked flap is lifted and folded again.
- **Coaching note:** get the seam straight down the middle before you reach for the tape.

Violation: Strip handled off the grab tab

- **Visible cue:** a gripper pinches the strip anywhere but its grab tab, touches the adhesive face, or reaches inside the tape dispenser instead of taking the presented tab.

- **SOP rule broken:** Step 4, the grab tab is the only part of a strip a gripper touches.
- **Coaching note:** take only the tab presented by the dispenser.

Violation: Strip laid off the seam or reused after a bad landing

- **Visible cue:** the strip lands at an angle, off to one side, short of either end, or folded onto itself, and is reused instead of being placed on the tape scrap rest.
- **SOP rule broken:** Step 4, lay the strip along the whole seam; scrap a damaged or badly laid strip and take the next strip presented by the dispenser.
- **Coaching note:** tape sticks once. Move the bad strip to the scrap rest and take the next one.

Violation: Seam not pressed

- **Visible cue:** the strip is laid but never pressed, pressed only at one end, pressed in several separate dabs, or left with an end lifting or a bubble standing up when the carton moves.
- **SOP rule broken:** Step 4, press along the strip from far to near in one stroke without coming off the tape until it lies flat with no lifted end.
- **Coaching note:** one stroke, all the way across, without leaving the tape.

Violation: Carton left off the finished spot

- **Visible cue:** the sealed carton stops short of the finished spot, overshoots it, stands turned, or is left standing on the stuffing spot at the end of the slide.
- **SOP rule broken:** Step 5, slide the carton left until it stands square and centered on the finished spot with its facing unchanged.
- **Coaching note:** take it all the way onto the spot and square, in one slow slide.

Violation: Turned in the air

- **Visible cue:** the flyer, the sample, or the strip is rotated or flipped between lift and placement, or the carton turns during the guided slide.
- **SOP rule broken:** Steps 1 to 5, held items keep their staged orientation, and the carton keeps its facing while sliding on the table.
- **Coaching note:** preserve orientation during picks and use the guide gripper during the carton slide.

Violation: Held item dragged instead of lifted

- **Visible cue:** the flyer, the sample, or the strip drags along its spot during a pick instead of lifting clear. The prescribed carton slide, flap folds, and tape draw are not violations.
- **SOP rule broken:** Steps 1 and 4, lift a held item clear before carrying it sideways.
- **Coaching note:** lift held items; slide only the carton as prescribed.

Violation: Regripped in the air

- **Visible cue:** a gripper shifts, rolls, or seats its hold again on the flyer, the sample, or the strip without setting it down first. Travelling contact against a carton wall during the slide is not a grasp.
- **SOP rule broken:** Steps 1 and 4, set a held item down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the flyer spot, the sample spot, the tape dispenser, the finished spot, or open table.
- **SOP rule broken:** Steps 1 to 5, a gripper opens inside the carton, on the tape scrap rest, or after the strip is resting on the seam. The carton is moved with closed grippers.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** the flyer or the sample is nudged after release, the sealed top is pushed to square a flap, or the carton is corrected with small pats instead of being returned to the start of its slide.
- **SOP rule broken:** Steps 1 to 5, lift and replace a bad insert, fold a bad flap again, and return a misplaced carton to the slide's starting point before repeating the full slide.
- **Coaching note:** restart the prescribed motion instead of making small corrective shoves.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** the flyer, the sample, or the strip falls short of its destination; the sample topples; the carton tips; the dispenser, scrap rest, flyer spot, or sample spot shifts; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 5, keep each held item on a clear travel path, the carton on its base, and both arms clear of each other.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the carton open, unsealed, or still on the stuffing spot, the flyer or the sample left on its spot, a strip stuck to a gripper, an arm away from home, an arm still over the finished spot, or recording stopped before 3 seconds of a still scene.
- **SOP rule broken:** Step 6, confirm the sealed carton on the finished spot and the empty flyer spot, sample spot, and stuffing spot; leave the scene untouched for 3 seconds; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm and hold first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective carton:** a wall that crushes under a correct push, a base that catches or tips on a clean table, a flap that tears at its fold, or bottom tape that lets go. Replace it before the next episode.
- **Defective flyer:** creased, torn, or curled so it will not lie flat on the carton floor. Replace it before the next episode.
- **Defective sample:** one that will not stand on its own on a flat carton floor, or whose cap comes off in the gripper. Replace it before the next episode.
- **Tape dispenser fault:** it fails to present a strip, cuts a strip shorter than the seam, presents a tab with adhesive on it or out of the gripper's reach, or does not advance after a correct pick. Service it before the next episode.
- **Stuffing spot, finished spot, dispenser, or tape scrap rest shifts out of its zone** during the episode.
- **Carton slides** under a correct flat hold because the stuffing spot is dusty or slick. Wipe it before the next episode.

Annotation subtasks (from SOP)

1. Lower the flyer into the carton face up
2. Lower the sample into the carton upright
3. Hold the open carton still to confirm both inserts
4. Fold the near and far top flaps in
5. Fold the left and right top flaps in
6. Lay one tape strip along the top seam
7. Press the seam strip flat
8. Slide the sealed carton onto the finished spot
9. Return both arms home and end the episode